NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
MOOCs; July - Dec 2026
Indian Institute of Technology Guwahati
Assignment No. 3: Detailed Solution

10 Marks

Each question carries 01 Mark each.

Q1. Which ONE of the following statements is false w.r.t AND/OR Graphs?

A. A solution graph is analogous to a path in an ordinary graph.

B. Every OR node contributes exactly one selected successor to the solution graph.

C. Every AND node contributes all of its successors to the solution graph.

D. AO* always expands every node of the graph before terminating.

Correct Answer: (D)

Explanation: AO* is a best-first heuristic algorithm. It expands only the most promising partial solution graph and can terminate as soon as the best solution graph is found, leaving many nodes unexpanded.

Q2. Suppose an OR node has three successors. Assume the heuristic values represent the estimated remaining cost from each successor to a goal state.

Node	Edge cost	Heuristic
A	2	6
B	1	8
C	4	2

Which node is selected first by AO*?

A. Node A

B. Node B

C. Node C

D. Cannot be determined

Correct Answer: (C)

Explanation:

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f(A) = 2 + 6 = 8
f(B) = 1 + 8 = 9
f(C) = 4 + 2 = 6

AO* selects node with minimum f -> C (f = 6)
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Q3. In game playing, which assumption distinguishes Minimax from ordinary shortest-path search?

A. The graph is cyclic.

B. The opponent always plays optimally.

C. Edge costs are negative.

D. The heuristic is admissible.

Correct Answer: (B)

Q4. A game such as Chess has a large branching factor and a very deep game tree.
Which of the following are correct?

I. Exhaustive search to terminal positions is impractical because the number of leaf nodes grows exponentially with search depth.
II. Limiting the search depth and applying a heuristic evaluation function at non-terminal nodes provides an estimate of the game state's utility.
III. Alpha-Beta pruning changes the Minimax decision obtained at the root by eliminating inferior branches.
IV. The final decision is based on backed-up utility estimates rather than exact game outcomes.

A. I and II only

B. I, II and IV only

C. II, III and IV only

D. All four statements

Correct Answer: (B)

Q5. Which of the following statements are true for searching an AND-OR graph?

I. To find solution in an AND-OR graph we need an algorithm similar to best-first search; but with the ability to handle the AND arc appropriately.
II. Algorithm should find a path from the start node to a set of nodes representing the solution states.

A. Both I and II
B. Neither I nor II
C. I Only
D. II Only

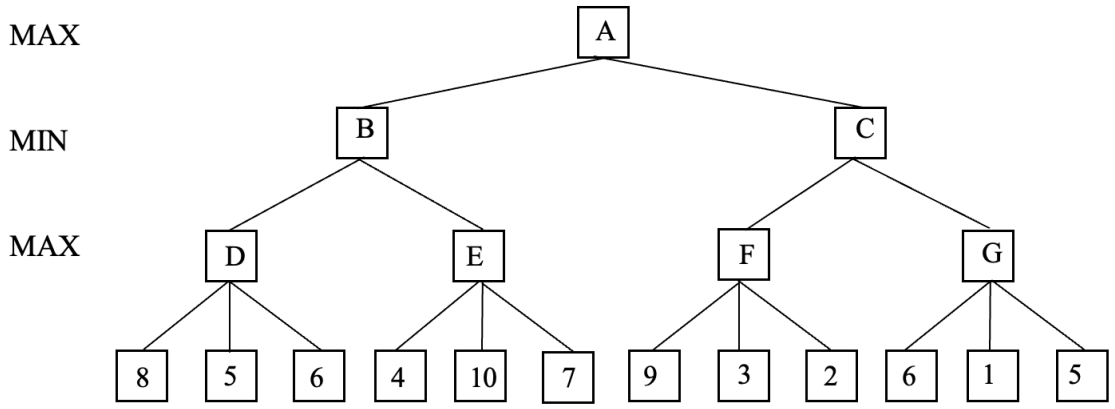
Correct Answer: (A)

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For Q6 – Q10

Consider the following game tree. The root node is a MAX node, and the levels alternate between MIN and MAX nodes. Terminal nodes contain the utility values shown. During Minimax and Alpha-Beta search, child nodes are evaluated from left to right.

Based on the game tree, answer Questions 6–10.



Based on the above data, answer the following questions.

Q6. After the complete evaluation of node B, what are the values of α and β at the root node A?

A. $\alpha = 6, \beta = +\infty$
B. $\alpha = 8, \beta = +\infty$
C. $\alpha = 10, \beta = +\infty$
D. $\alpha = 8, \beta = 8$

Correct Answer: (B)

Explanation: Minimax value of every node: D = 8, E = 10, B = 8, F = 9, G = 6, C = 6, A = 8.

Alpha – Beta root value = 8.

Leaves examined (in order): 8, 5, 6, 4, 10, 9, 3, 2, 6, 1, 5.

After evaluating B: at root A: alpha = 8, beta = infinity.

Q7. During left-to-right Alpha-Beta search, the first pruning occurs during the evaluation of which node?

A. Node D
B. Node E
C. Node F
D. Node G

Correct Answer: (B)

Explanation: First pruning happens at node E.

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Q8. Which one of the following leaf values is never examined during Alpha-Beta search?

A. 10
B. 7
C. 1
D. 5

Correct Answer: (B)

Explanation: Only the cut-off at E skips a leaf, and that leaf is the third child of E (value 7). Every other leaf is read.

Q9. During Alpha-Beta search, suppose node D has been evaluated but node E has not. Which of the following is true?

A. α at node B is 8.
B. β at node B is 8.
C. β at the root is 8.
D. α at the root is 8.

Correct Answer: (B)

Explanation: At node B, after child D and before E: alpha(B) = - infinity, beta(B) = 8.

Q10. Suppose the move ordering at every MAX node changes from best-first to worst-first. Which of the following is correct?

A. More nodes are likely to be evaluated.
B. Fewer nodes are guaranteed to be evaluated.
C. The Minimax value changes.
D. Alpha-Beta pruning becomes incorrect.

Correct Answer: (A)

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